

## Extra credit problems

Math 427

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)

1. Describe all the motions of the Manhattan plane.

2. Construct a metric space  $\mathcal{X}$  and a distance-preserving map  $f: \mathcal{X} \rightarrow \mathcal{X}$  that is not a motion of  $\mathcal{X}$ .

3. Note that the following quantity

$$\tilde{\angle} ABC = \begin{cases} \pi & \text{if } \angle ABC = \pi, \\ -\angle ABC & \text{if } \angle ABC < \pi. \end{cases}$$

can serve as the angle measure; that is, the axioms hold if one changes everywhere  $\angle$  to  $\tilde{\angle}$ .

Show that  $\angle$  and  $\tilde{\angle}$  are the only possible angle measures on the plane, but without Axiom IIIc, this is not longer true.

4. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.